VALIDATION REPORT

Next-Generation Catastrophe Model: Loss Output Fitness for Client Use

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Executive summary

Recommendation: REJECT. Not fit for client use in its current form.

1. Dataset contains data quality defects that should be fixed.
2. Event frequency does not decline with intensity and does not vary by geography.
3. Event vulnerability are not differentiated by event type.

The dataset is assessed according to three broad buckets consisting of data quality/integrity, event intensity and modelled loss checks.

Data set summary:

- 1,057,320 event location loss records
- 73,824 loss causing events (wind, hail, tornado)
- ~10,000 catalog years
- 243,606 locations

Data Integrity Checks

Certain data quality issues were detected in the dataset:

1) Geographic issues
    a. All 8,722 locations with ZIP 320–349 and FL coordinates carry state code CA; FL is absent.
    b. 51 locations are offshore of Maine or north of 49°N.
2) 4,130 events (6,286 rows) are assigned to more than one catalog year.
3) Two defect populations of exactly the same size (3,172 null losses and 3,172 "_ERR" years) that never overlap.
4) 13,307 wind rows have intensity 0–5 against a valid range of 50–180, with an empty gap between. Their losses look normal.
5) All three event types affect locations within the same ~16 km radius and have indistinguishable distance distributions. (Tornado damage paths are narrow tracks, while severe wind systems extend over regional scales)

Assumptions:

- Units for Wind, hail and tornado intensity are miles per hour, inches, EF Scale.
- 10,000 year stochastic catalog

Event Intensity Checks

1) Frequency (Pass)

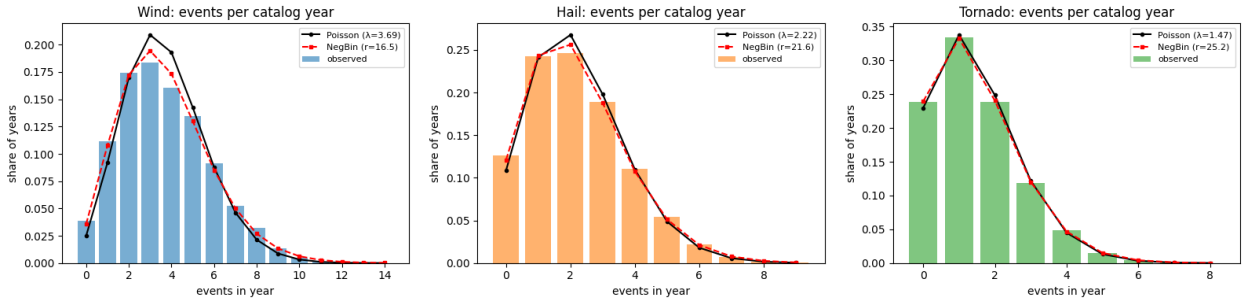


Figure 1. Distributions of events per catalog year for each event type. Poisson and Negative Binomial fitted to each Distribution

2) Frequency vs intensity (Fail)

- For wind the relationship is inverted.
- For hail and tornado, event counts are uniform across the intensity range.
- Observed climatology falls steeply with severity for example, EF4–EF5 tornadoes are on the order of 1% of US tornadoes.

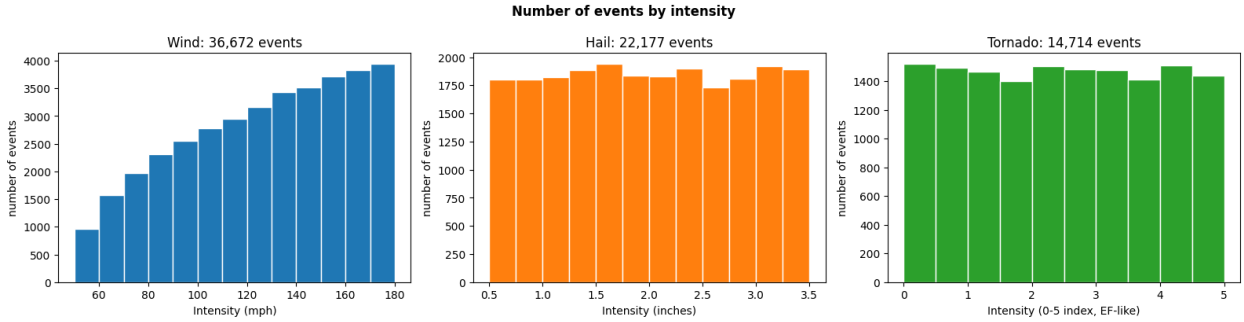


Figure 2. Event frequency vs intensity plotted for all event types.

3) Geographic variation (Fail)

Even rates vary across states by a coefficient of variation of only 8–10%. A location in Oklahoma is hit by a tornado about as often as one in Washington. The model does not reproduce the regional hazard patterns (Tornado Alley, hail belt, coastal wind gradient).

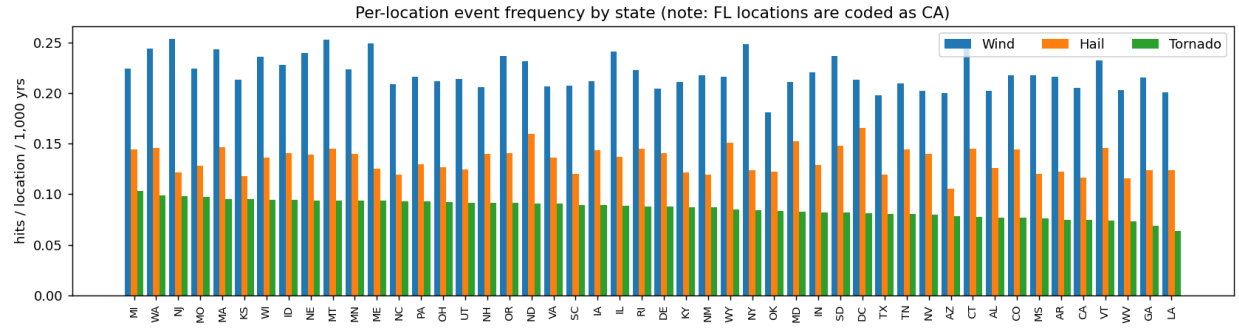


Figure 3. Per-location event frequency by state and event type, sorted by tornado frequency.

Modelled Loss Checks

Distribution (Pass)

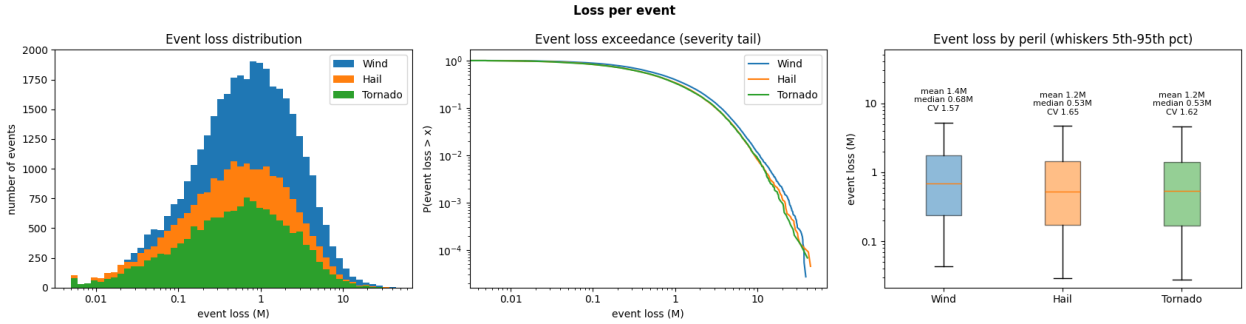


Figure 4: Losses in millions for each event type are normally distributed

Vulnerability (Fail)

Loss increases with intensity (Spearman ρ 0.55–0.67) and decays with distance from the event center (ρ −0.57 to −0.64). However, after rescaling each event's intensity to a common 0-1 range, the hail and tornado loss curves agree within 2% (Figure 5, left). This indicates one generic damage function reused across event types.

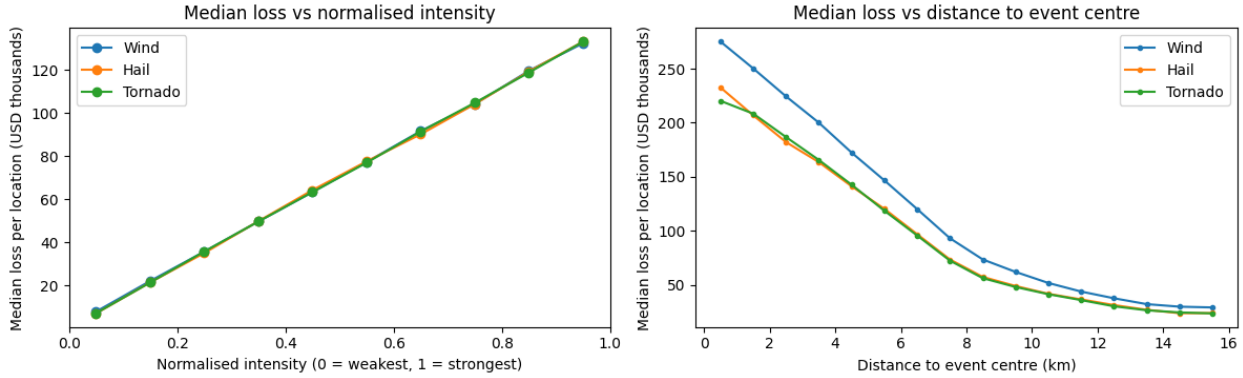


Figure 5. Median loss by normalized intensity (left) and by distance from event center (right).

Exceedance Probability EP (Sanity Check?)

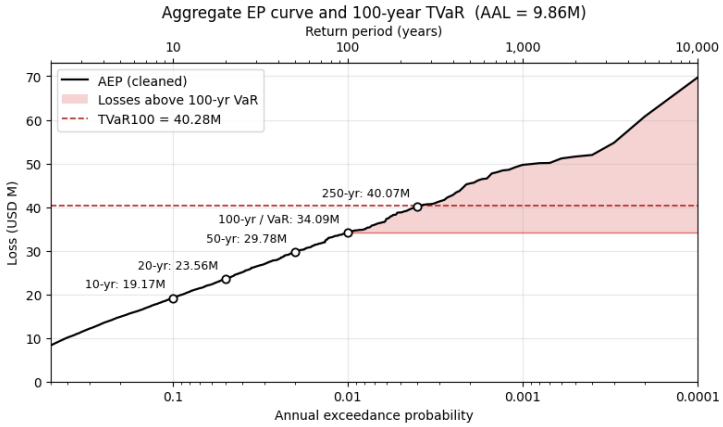


Figure 6: Plot of Losses against exceedance probability to compare expected losses for Value at Risk.

Additional analyses/checks with more time

- Sanity checks against actual data for AAL, OEP, AEP, VaR, TVaR
- Create dashboard from workflow to analyze future files.
- Quantitative benchmarking of event frequency by state and magnitude against datasets (eg. NOAA SPC storm reports).
- Comparison of modelled losses with reported industry losses for historical events.
- Tests of temporal clustering and seasonality.

References

1) www.verisk.com/blog/managing-severe-thunderstorm-risk-iii/
2) descartesunderwriting.com/insights/understanding-global-weather-perils-tornado-risk